

Arduino UNO Based LED Indicator Model – Detailed Report

1. Introduction

This project titled “Ingenious Tumelo-Fulffy” is an Arduino UNO based LED indicator model designed using Tinkercad. The model demonstrates the fundamental working of a microcontroller by controlling multiple LEDs connected through current limiting resistors. It is mainly intended for academic demonstration, mini-projects, and electronics learning purposes.

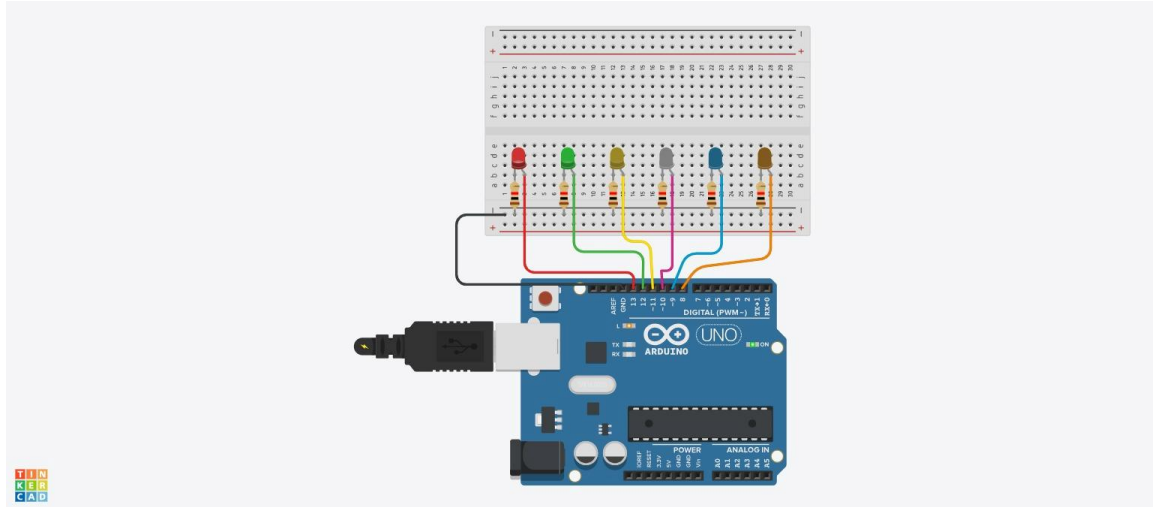
2. Objective of the Model

The main objective of this model is to understand:

- Digital output control using Arduino UNO
- Interfacing LEDs with a microcontroller
- Importance of resistors in current limitation
- Pin configuration and power distribution in Arduino boards

3. Block Diagram Description

The Arduino UNO acts as the control unit of the system. Digital pins D2 to D6 are used as output pins. Each pin is connected to an LED through a 1kΩ resistor. The USB cable supplies 5V power to the Arduino board. Ground (GND) is common for all LEDs, completing the circuit.



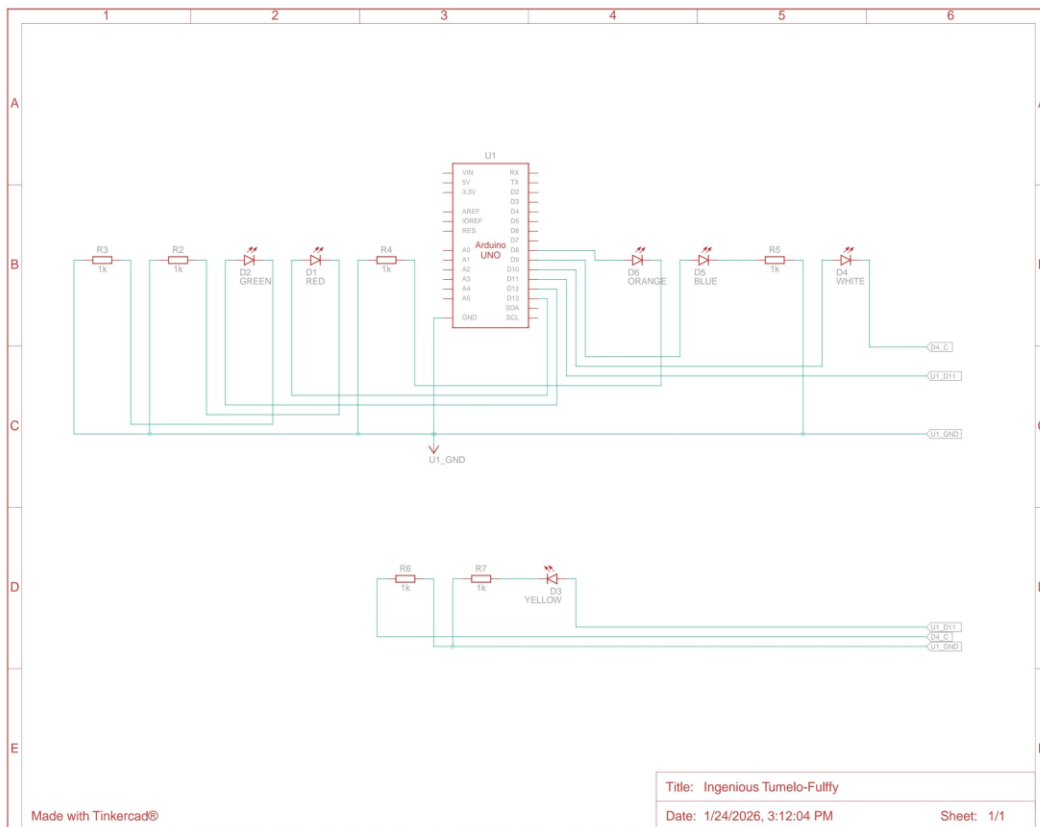
4. Working Principle

When power is supplied to the Arduino UNO through a USB cable, the microcontroller executes the uploaded program. The program configures digital pins D2, D3, D4, D5, and D6 as output pins. The Arduino sends HIGH signals sequentially to these pins, causing the corresponding LEDs (Red, Yellow, White, Blue, and Orange) to glow. After a short delay, the pins are set to LOW, turning OFF the LEDs. This process repeats continuously, creating an LED sequence pattern.

5. Circuit Description

According to the provided Tinkercad circuit diagram:

- D2 pin is connected to a Green LED through a $1k\Omega$ resistor
 - D3 pin is connected to a Yellow LED through a $1k\Omega$ resistor
 - D4 pin is connected to a White LED through a $1k\Omega$ resistor
 - D5 pin is connected to a Blue LED through a $1k\Omega$ resistor
 - D6 pin is connected to an Orange LED through a $1k\Omega$ resistor
- All LED cathodes are connected to the GND pin of the Arduino UNO.



6. Components Used

S.No	Component	Type / Model	Quantity	Capacity / Rating
1	Microcontroller Board	Arduino UNO R3	1	5V operating voltage
2	LED – Red	5mm LED	1	Forward Voltage: ~2V
3	LED – Green	5mm LED	1	Forward Voltage: ~2.2V
4	LED – Yellow	5mm LED	1	Forward Voltage: ~2.1V
5	LED – Blue	5mm LED	1	Forward Voltage: ~3.2V
6	LED – White	5mm LED	1	Forward Voltage: ~3.2V
7	Resistors	Carbon Film Resistor	5	1k Ω , 0.25W
8	Jumper Wires	Male-to-Male	As required	5V compatible
9	USB Cable	USB Type-B	1	5V DC supply

7. Advantages

- Simple and easy to understand
- Low cost and reusable components
- Safe low-voltage operation
- Ideal for beginners in embedded systems

8. Applications

- LED indicator panels
- Traffic signal demonstration models
- Educational Arduino experiments
- Sequential lighting systems

9. Circuit Diagram Explanation

The circuit diagram for this project is designed using Tinkercad and consists of an Arduino UNO board, five LEDs, and current-limiting resistors. The Arduino UNO acts as the central control unit.

Each LED is connected to a specific digital output pin of the Arduino UNO through a 1k Ω resistor. The resistor is used to limit the current flowing through the LED, thereby preventing damage. The anode (positive terminal) of each LED is connected to the respective digital pin, while the cathode (negative terminal) is connected to the common ground (GND).

Pin connections as per the circuit diagram:

- Digital pin D2 → Green LED via 1k Ω resistor
- Digital pin D3 → Yellow LED via 1k Ω resistor
- Digital pin D4 → White LED via 1k Ω resistor
- Digital pin D5 → Blue LED via 1k Ω resistor
- Digital pin D6 → Orange LED via 1k Ω resistor

The Arduino UNO receives power from a USB cable, which supplies 5V DC. All components operate at safe low voltage, making the circuit reliable and suitable for educational purposes.

10. Arduino Program Explanation

The Arduino program controls the LEDs by sending digital HIGH and LOW signals to the output pins. The logic of the program is simple and easy to understand.

Initially, the required digital pins (D2 to D6) are declared as output pins using the `pinMode()` function inside the `setup()` section. The `setup()` function runs only once when the Arduino is powered ON or reset.

The `loop()` function runs continuously. Inside this function, each LED is turned ON by setting the corresponding pin to HIGH using `digitalWrite()`. A delay is introduced using the `delay()` function to keep the LED ON for a short duration. After the delay, the pin is set to LOW, turning the LED OFF. This process repeats for all LEDs in sequence, creating a running light effect.

This program demonstrates important Arduino concepts such as digital output control, timing using delays, and sequential execution of instructions.

11. Conclusion

The Arduino UNO based LED indicator model successfully demonstrates digital output control using a microcontroller. By referring to the Tinkercad circuit diagram, the model shows proper pin configuration, component selection, and safe operation. This project serves as a strong foundation for advanced automation and embedded system applications.

<https://www.tinkercad.com/things/4kX4P3dDs24-ingenious-tumelo-fulffy>